GEK-83846

DRIVE SYSTEM INSTRUCTIONS

POWER CONVERSION MODULE
1000 HP AT 600VAC
DS3820PCD

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 241 53

GENERAL ELECTRIC

SECTION 6 - 1

POWER CONVERSION MODULE

1000 HP AT 600VAC

DS3820PCD

(GEK - 83846)

FUNCTION

The power conversion module converts a constant frequency AC voltage into a variable DC voltage that can be used by a DC motor. The amount of power that the power conversion module delivers to the motor is controlled by the Propulsion Drive Module (PDM).

OPERATION

The power conversion module consists of 3 major parts:

- The SCR Bridge
- Cooling System
- Control or High Voltage Interface

The SCR Bridge contains the SCRs (Silicon Controlled Rectifiers) which will conduct current by precisely controlled gating signals. Thus a variable DC voltage is produced, which controls the motor speed. In addition, the Bridge contains anode to cathode snubbers, and line/leg reactors. The snubbers limit the rate of use of voltage and the reactors limit the rate of use of current within the design limits of the SCRs.

Heat generated by the SCR bridge during operation is removed via the heat sinks. Two cooling fans mounted below the bridge force air over the heat sinks to remove the excess heat. The drive control module monitors the cooling fans by means of two printed circuit boards located on the front of the fan assembly. These fan loss cards can detect that a fan has failed and notify the PDM to shut down the SCR bridge.

The High Voltage Interface consists of a printed circuit card. Its purpose is to scale down high voltage and current to low level signals for use in the control electronics. In addition, temperature sensors are mounted on this card to monitor the ambient temperature of the bridge and send a signal proportional to temperature to the control electronics (PDM).

REMOVAL AND REPLACEMENT PROCEDURES

Removal and Replacement of PCM Cell Stack and Fuses

REMOVAL:

1. Remove all power from the panel. Open the panel door and then open the power control module door. Remove the 3 clips that latch the hinged glastic baffle.

2. Remove the ribbon cable clamp from the lower left hand sidewall; (All models do not have this clamp.)

3. Open the hinged glastic baffle as far foward as possible. Do not force.

4. Remove two screws at the top corners of the flat glastic baffle. Lift the right hand side and move it toward the front of the module, leaving the left hand side nearly in place. Slide two grommets out of the notches in the left hand edge and remove the baffle.

5. Remove four sets of bolts, washers, and lockwashers retaining the two fuses and remove the fuses. Lay spacers with wires attached to one side.

   NOTE: If only fuses require replacement, proceed to REPLACEMENT procedure at this point.

6. Gripping the "ears" of the Mate-n-Lock plug, squeeze inward and pull upward on the plug at the top and downward on the plug at the bottom to disconnect the module harness from the gate leads.

7. Remove the nut, washer, and lockwasher holding the bottom of the stack assembly.

8. While holding the stack assembly in place by hand, remove the bolt, washer, and lockwasher holding the top of the stack assembly. This releases the stack assembly for removal.

9. Move the stack assembly toward the front of the module carefully, being aware of the "stud" in the bottom hole. Remove the assembly from the module.

REPLACEMENT:

1. Clean mating surfaces of the replacement heat sink with #360 or finer sandpaper. Apply #2 EJC grease or equivalent, wipe clean, and apply a fresh coat.

REPLACEMENT (Continued):

2. Reassemble in reverse order of the removal procedure described above; Torque stack assembly and fuse mounting hardware to 12.5 foot-pounds; Glastic baffles must be replaced exactly as they were to prevent possible overheating of the cells;

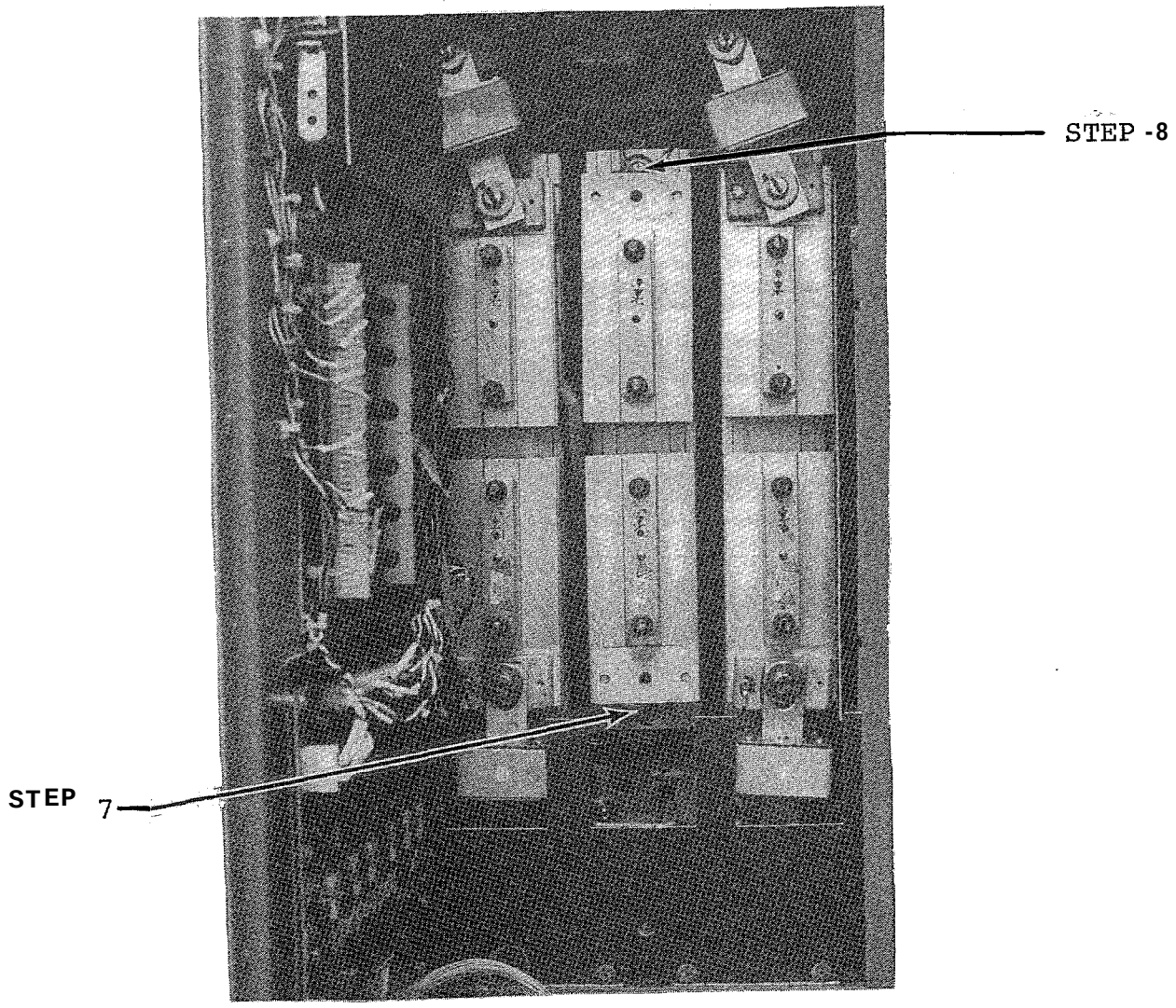


Figure 1 - PCM CELL STACK ASSEMBLY READY TO BE REMOVED

REPLACEMENT (Continued):

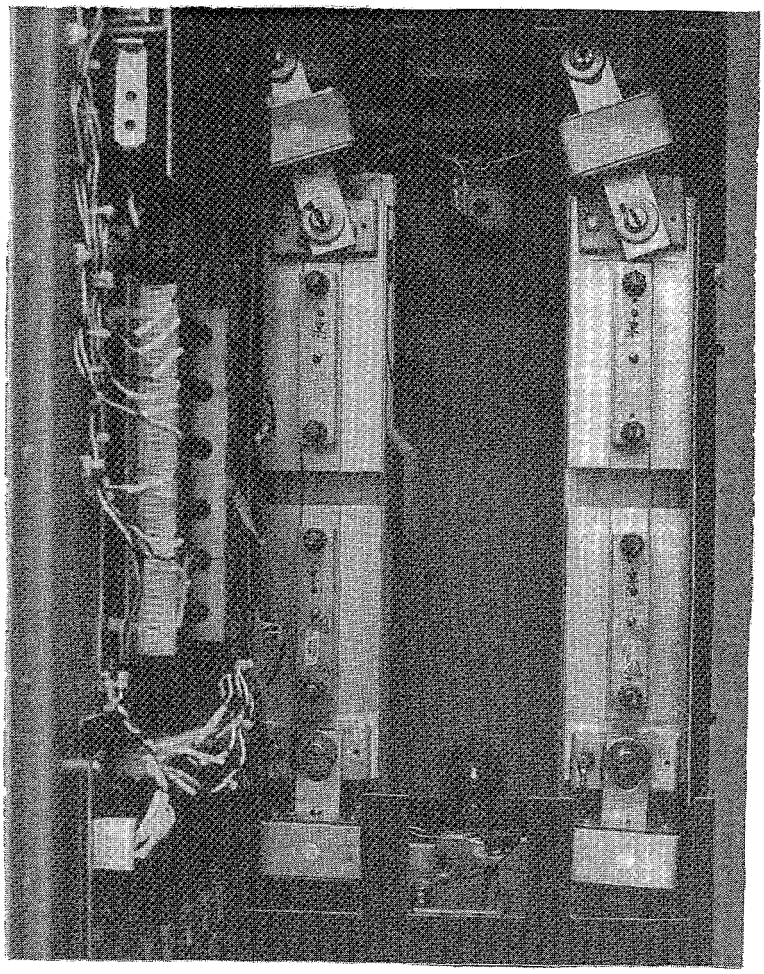


FIGURE 2 - PCM WITH ONE CELL STACK ASSEMBLY REMOVED

Removal and Replacement of PCM Fans

Refer to Figure 3 during the following procedures.

REMOVAL:

1. Remove all power from the panel. Open the PCM door and remove three screws at the bottom front edge. Unplug the wire snap-lock connectors.

REMOVAL (Continued):

2. Grasp the fan assemlby and firmly pull toward the front of the enclosure. Remove the fan assembly and place it on a table.

3. At this point there are three options:

   A. Replace the entire fan assembly by reversing the above instructions.

   B. Determine which fan is faulty and replace it using the following procedure:

      (1) Disconnect the random wire plug containing red and black wires from the fan loss card.

      (2) Using a semi-pointed instrument, gently depress the retaining clip of the two wire terminals located on the nonlocking side of the plug and remove the terminals from the plug housing.

      (3) Remove the four fan retaining screws from the fan housing (motor and blade only).

      (4) Remove the wire stress relief from the housing and lift out the fan while feeding the wires through the hole in the housing.

   C. Determine which fan is faulty and replace it using the following procedure:

      (1) Disconnect the random wire plug containing red and black wires from the fan loss card.

      (2) Using a semi-pointed instrument, gently depress the retaining clip of the two wire terminals located on the nonlocking side of the plug and remove the terminals from the plug housing.

      (3) Drill and remove the four blind rivets securing the fan and remove the fan.

REPLACEMENT:

Replace the fans by reversing the order of the removal procedure you used. If procedure C(3) was used, you can replace the blind rivets with bolts, nuts, and lock washers of the equivalent size.

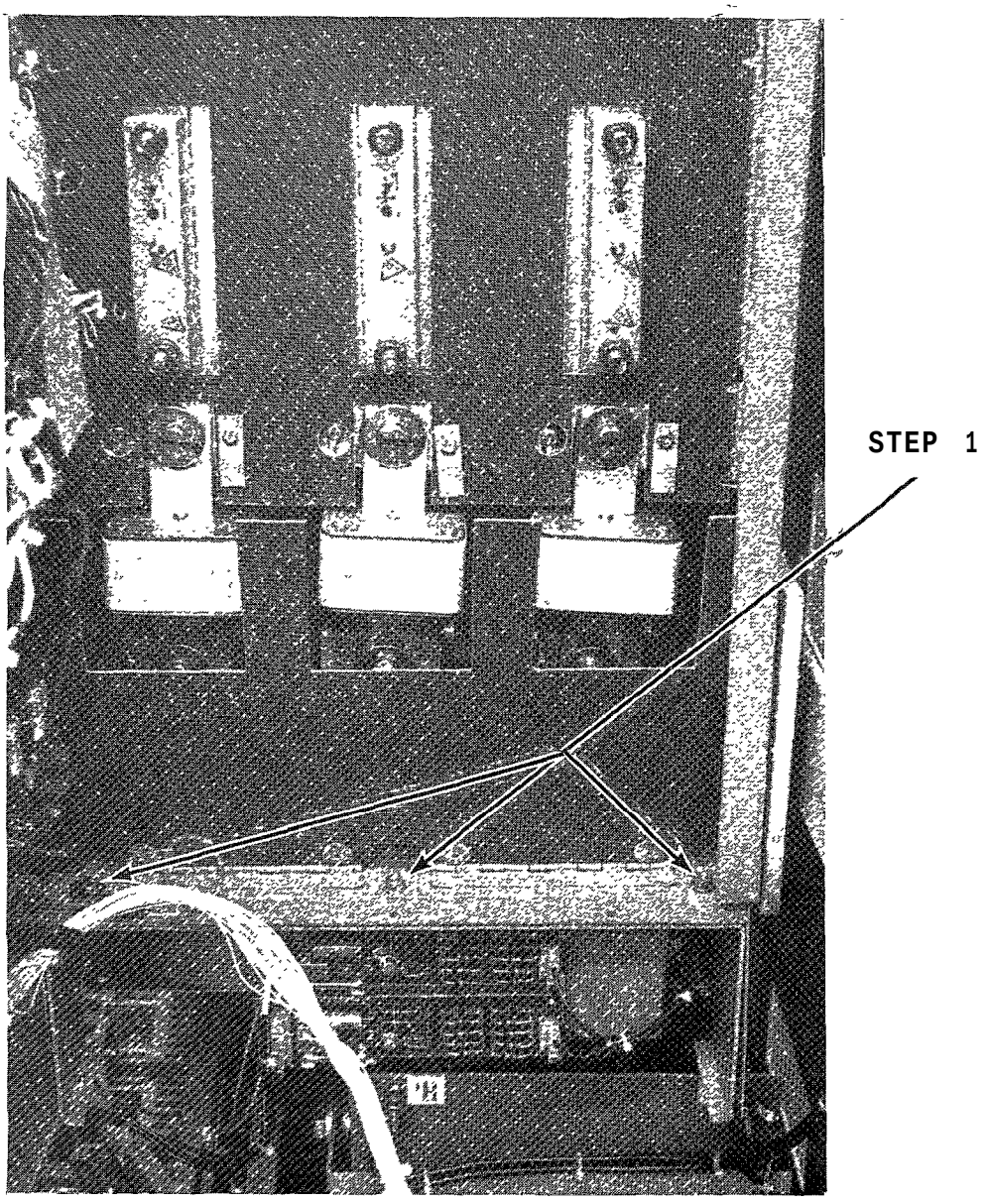
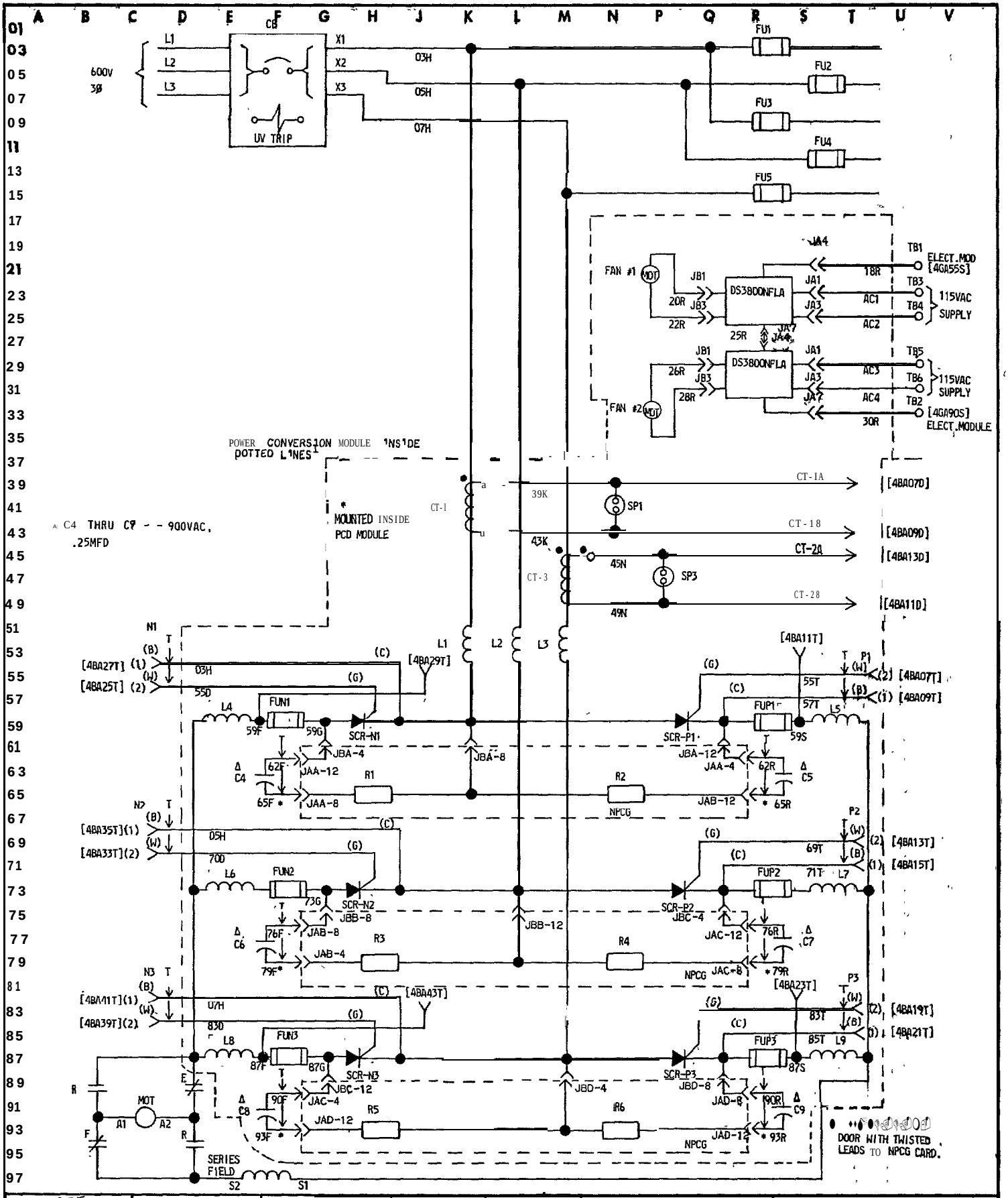


FIGURE 3 - PREPARING TO REMOVE PCM FAN ASSEMBLY

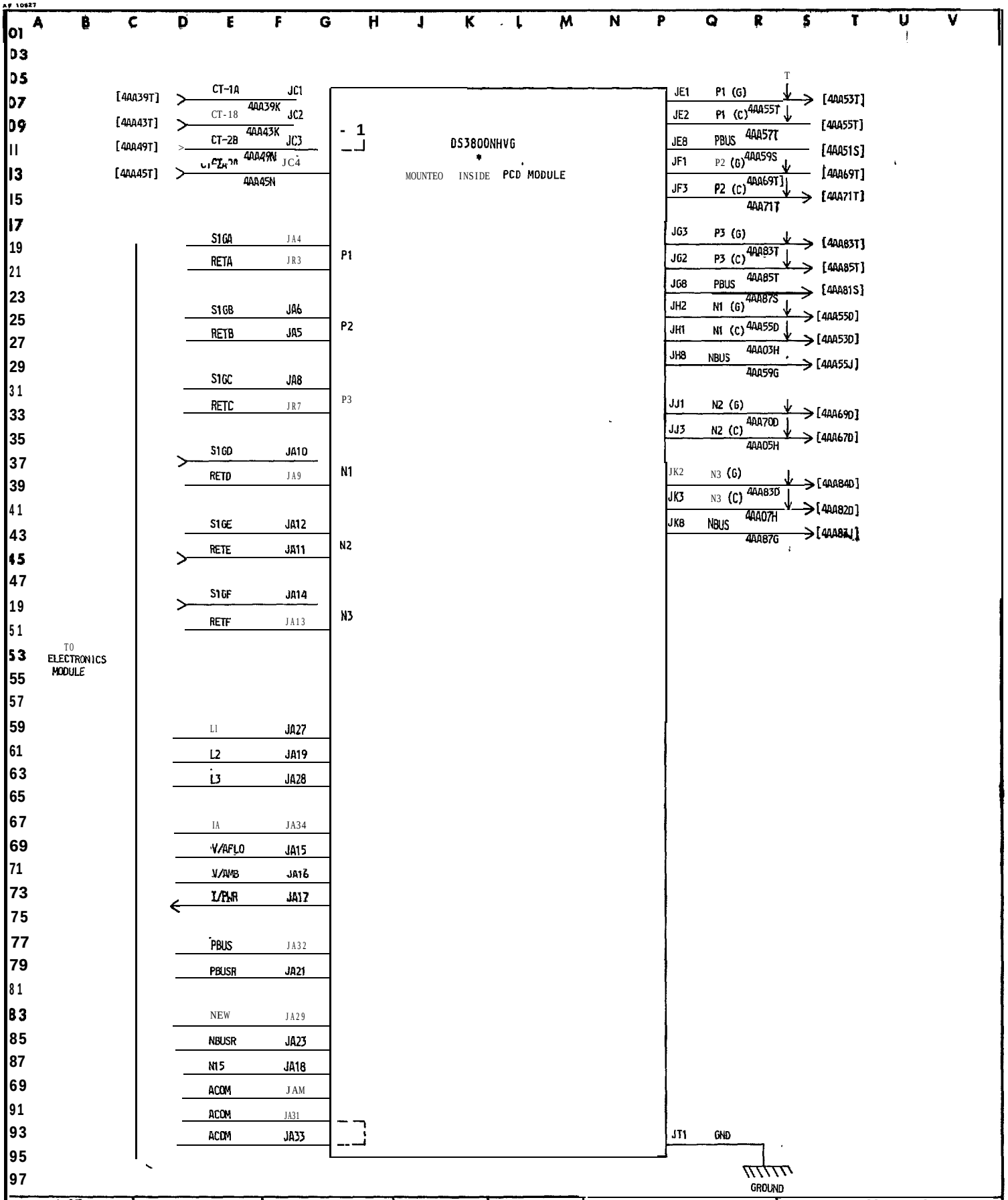
ELECTRICAL DIAGRAMS

The remainder of this tab section contains electrical diagrams of the power conversion panel.



REV. 1 OEE 8-3-81 CCH	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2 REVISED PLUG POINTS JA8 JB CCH 9-7-82	ISSUED 9/26/80				DRIVE SYSTEMS DEPARTMENT	OIL WELL DRILLING
REV. 3	MADE BY F. BARTLEY				SALEM, VA. U.S.A.	POWER BRIDGE
						DS3820PCD
						CONT. ON SH. 4BA SH. NO. 4AA

CARS



REV. 1 OEE 8-3-81 CCH	REV. 4	REV. 6	PRINTS TO DL114	ENGINEER M. Jarvinen	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
REV. 2	ISSUED 9/26/80				
REV. 3	MADE BY F. BARTLEY				

ELEMENTARY DIAGRAM

OIL WELL DRILLING
HIGH VOLTAGE BOARD

DS3820PCD

CONT. ON SH. 4 CASH. NO. 4BIA

CARS

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC